

Assignment 8 PROBABILITY vs STATISTICS

Each complete question and its options are followed by the correct answer.

Question 1: You know that a fair die has six equally likely outcomes. You want to calculate the probability of getting a 6. This is an example of: A. Statistics B. Probability C. Regression D. Sampling

Answer: B. Probability

Question 2: You roll a die 1,000 times and record the results. You notice that the number 6 appeared much more frequently than expected. You want to determine whether the die may be biased. This is primarily: A. Probability B. Statistics C. Linear Algebra D. Calculus

Answer: B. Statistics

Question 3: Which statement best describes Probability? A. Using observed data to estimate unknown characteristics B. Using a known model or assumptions to reason about possible outcomes C. Organizing data into tables D. Calculating averages from collected data

Answer: B. Using a known model or assumptions to reason about possible outcomes

Question 4: Which statement best describes Statistics? A. Predicting possible outcomes from a known probability model B. Using observed data to understand or estimate an underlying process C. Generating random numbers D. Calculating only probabilities between 0 and 1

Answer: B. Using observed data to understand or estimate an underlying process

Question 5: A company knows that historically 2% of its products are defective. It wants to estimate how many defective products might appear in the next 10,000 products. This is primarily: A. Statistics B. Probability C. Data cleaning D. Visualization

Answer: B. Probability

Question 6: A company collects data from 10,000 products and finds that 250 are defective. It wants to estimate the defect rate of the production process. This is primarily: A. Probability B. Statistics C. Linear Algebra D. Optimization

Answer: B. Statistics

Question 7: Which direction best represents Probability? A. Data → Model B. Model → Possible outcomes C. Data → Visualization D. Sample → Database

Answer: B. Model → Possible outcomes

Question 8: Which direction best represents Statistics? A. Model → Possible outcomes B. Possible outcomes → Model C. Observed data → Understanding/inference about the underlying process D. Probability → Randomness

Answer: C. Observed data → Understanding/inference about the underlying process

Question 9: A medical researcher knows that a disease occurs in approximately 1% of the population. She wants to calculate the probability that a randomly selected person has the disease. This is primarily: A. Statistics B. Probability C. Data preprocessing D. Regression

Answer: B. Probability

Question 10: A medical researcher observes that 15 out of 1,000 randomly selected people have a particular disease. She wants to estimate the disease prevalence in the larger population. This is primarily: A. Probability B. Statistics C. Matrix algebra D. Optimization

Answer: B. Statistics

Question 11: Consider the following situation: "A coin is believed to be fair. What is the probability of getting 7 Heads when the coin is tossed 10 times?" This is: A. Statistics B. Probability C. Descriptive statistics D. Sampling

Answer: B. Probability

Question 12: Now consider: "A coin is tossed 10 times and produces 9 Heads. Based on this observation, determine whether the coin is likely to be fair." This is: A. Probability B. Statistics C. Matrix multiplication D. Data visualization

Answer: B. Statistics

Question 13: Which question is a Probability question? A. What is the average height of students in this class? B. What is the probability that a randomly selected student is taller than 180 cm? C. What is the median height of the students? D. What is the standard deviation of student heights?

Answer: B. What is the probability that a randomly selected student is taller than 180 cm?

Question 14: Which question is a Statistics question? A. What is the probability of getting Heads on a fair coin? B. What is the probability of rolling a 6? C. What is the average income observed in this dataset? D. What is the probability of rain tomorrow?

Answer: C. What is the average income observed in this dataset?

Question 15: A weather model says that there is a 70% probability of rain tomorrow. This statement is an example of: A. Statistics B. Probability C. Descriptive statistics D. Sampling

Answer: B. Probability

Question 16: You collect the rainfall measurements from the last 20 years and calculate the average annual rainfall. This activity is primarily: A. Probability B. Statistics C. Probability distribution D. Random sampling Just to make it clearer – Probability and statistics

Answer: B. Statistics

Question 17: "What is likely to happen?" A. Probability B. Statistics

Answer: A. Probability

Question 18: "What happened in the data?" A. Probability B. Statistics

Answer: B. Statistics

Question 19: "What can we learn about the population from our sample?" A. Probability B. Statistics

Answer: B. Statistics

Question 20: "If the assumptions are known, what outcomes should we expect?" A. Probability B. Statistics

Answer: A. Probability